FINANCIAL ANALYTICS

Time Series Analysis & Forecasting using ARIMA On Reliance Dataset .

Submitted by
GROUP -1

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EXECUTIVE SUMMARY

The following report is a financial time series analysis of Reliance Industries Ltd. (NSE: RELIANCE.NS) stock prices expressed in dollars daily from 7th September 2012 to 7th September 2025. The research aimed to use the ARIMA modeling method in its forecasting and to understand more about the statistical characteristics of financial data.

The breakdown was done systematically.

- **Data Collection and Diagnostics**

Yahoo Finance was used to get historical price information of Reliance on its daily closing prices. The first visualization showed that its prices had a great uptrend, especially since 2017, and that the volatility was visible around the key market events like the 2020 COVID-19 pandemic.

These trends have the implications that the raw price series is not stationary, as is the case with most financial assets.

- **Testing and Transformation of Stationarity.**

This was done using two statistical tests: the Augmented Dickey-Fuller (ADF) test and the KPSS test. Upon ADF test, the null hypothesis of unit root was not rejected (p=0.1128) and the null hypothesis of stationarity was rejected (p. This established the non-stationarity.

To overcome this, data were log transformed and differenced to get returns. In the case of returns, the two tests (ADF p=0.01, KPSS p=0.10) indicated that returns were stationary, and could be used to model ARIMA.

- **Model Identification & Selection.**

The plots of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) indicated that a small-lag ARMA-type model would be suitable. A number of candidate models were experimented and they were ARIMA (1,0,0), ARIMA (2,0,0), ARIMA(1,0,1), ARIMA(2,0,2) and so on.

Akaike Information Criterion (AIC) and Sum of Squared Errors (SSE) were used as model performance metrics. ARMA (2,2) (i.e. ARIMA (2,0,2)) was the best model as it had the lowest AIC (-17054.18) and passed the diagnostic tests.

- **In-Sample Forecasting & Accuracy.**

The data was divided into 90 percent training and 10 percent testing data. ARMA (2,2) was used to generate forecasts for the period under testing and compared them against actual

returns. The metrics of accuracy were very good, with the value of RMSE 0.0142 and the MAE 0.0103. Such small errors show that the model followed real numbers in a close way.

- **Out-of-Sample Forecasting**

The ARMA (2,2) model was re-estimated with the whole dataset to get the best fit model and 14 days predictions were obtained. Findings indicated that the expected returns in future are expected to be around zero with small confidence intervals, which is in line with the efficient market hypothesis where excess returns are very hard to predict.

- **Residual Diagnostics**

The residual analysis ensured that the errors were treated as white noise with the errors showing no significant autocorrelation (Ljung-Box test p=0.695). This justifies the statistical sufficiency of the selected model.

OBJECTIVES

The main aim of this project was to perform time series analysis and forecast to real world financial data with special interest in Reliance Industries Ltd (NSE: RELIANCE.NS). The research also sought to bring theory and practice together by applying the ARIMA (Autoregressive Integrated Moving Average) model to make predictions about financial forecasting.

In particular, the goals can be divided into the following parts:

To Collect and Prepare Data

- Obtain at least 10 years of daily closing price data of Reliance as of the Yahoo Finance.
- Clean the dataset and deal with the missing values and generate a useful time-series object in R.

To conduct initial Diagnostics.

Graphically investigate and examine the raw prices (close) to find a trend, cluster of volatility or seasonality.

To Test for Stationarity

Both Augmented Dickey-Fuller (ADF) test and KPSS test on the raw price series.

Both tests have clearly spelled out null and alternative hypotheses.

Decision on whether transformation (logarithmic, differencing) is necessary.

To Transform the Series.

Stabilize variance by using log transformation.

Use differencing to eliminate trends and transform the series into a stationary process which can be modeled using ARIMA.

To Find Candidate ARIMA Models, Identify the appropriate value of AR (p) and MA (q) by use of ACF and PACF plots of the stationary returns.

To Compare and Estimate Models.

Fit various ARIMA specifications (i.e. ARIMA (1,0,0), ARIMA(2,0,2), etc.) to the data.

Compare models in terms of Akaike Information Criterion (AIC) and Sum of Squared Erros (SSE).

Choose the most successful model.

To make In-Sample Forecasting and Accuracy Assessment.

Divide the dataset in training (90%), testing (10%).

Make predictions on the test set and measure the accuracy in terms of ME, RMSE, MAE, and MAPE.

Compare the predicted series and the actual series.

To perform Out-of-Sample Forecasting.

- Refit the adopted best model on the entire data.
- Generate 14-day forecasts (two trading weeks).
- Analyze the results of the plot and interpret the results.

Perform Residual Diagnostics.

- Use Ljung-Box test to make sure that the residuals are white noise.
- Ensure that the model is statistically sufficient.

To draw Insights and Learning Results.

- Overview of the ARIMA model performance in the financial forecasting.
- Consider real world problems (market efficiency, volatility clustering and the weaknesses of linear time-series models).
- Record the project learning outcomes; both technical and conceptual.

Overall, running ARIMA in R using the sample was not the point of this assignment, but rather to get the whole forecasting process with its end-to-end: data collection, diagnostics, model identification, testing, forecasting, and interpretation of the financial environment of Reliance Industries.

LIBRARIES USED

To perform the analysis effectively, few R libraries were used. The choice of each package was because it offers specialized tools in time series modeling, statistical tests and visualization. Each library will be described in detail below:

The quantmod (Quantitative Financial Modelling Framework)

- It is a model designed to support analysis of the causes and effects of market movements, and to apply this knowledge in real trading of financial assets like stocks or bonds.

Purpose: The purpose is that of quantitative financial analysis and trading.

Key Functions Used:

getSymbols() - to directly download historical stock data (in this case Reliance Industries) of Yahoo Finance.

Cl, Ad() - to obtain downloaded price data of closing or adjusted closing price.

Role in Project: The library enabled us to gain access to 13 years of daily Reliance stock data in a way that was not reliant on manual downloading, which made the workflow process reliable and reproducible.

tseries (Time Series Analysis and Computational finance)

Purpose: It is a basic statistical test and time-series model.

Key Functions Used:

- adf.test (): Graphic of Augmented Dickey-Fuller test to identify unit roots in a series (to determine whether it is stationary).
- kpss.test (): KPSS test of level stationarity (complementary to ADF).
- **Role in Project**: These are the tests which were at the heart of our stationarity diagnostics. The combination of them resulted in a statistical finding that the raw prices of Reliance were non-stationary, but the returns were stationary.
- **forecast**
- **Purpose**: End to end prognostics and time-series prognostics.
- **Key Functions Used:**
- **autoarima ()- gets to the best ARIMA model.**

- **predictor -- produces projections of the future with projections.**
- **accuracy () - calculates the error measures including ME, RMSE, MAE, and MAPE.**

Role in Project: This library allowed us to fit ARIMA/ARMA models, create forecasts, view them with confidence intervals and calculate error measures to compare performance.

TTR (Technical Trading Rules)

Purpose: Implements technical analysis functions, which are common in finance.

Commonly Used Functions: Moving averages (SMA, EMA), Bollinger Bands, and volatility.

Role in Project: Although our focus was on ARIMA modelling, TTR assisted with exploratory analysis, e.g. by smoothing functions (moving averages), which revealed underlying trends of prices and aided in visual diagnostics.

lubridate

Purpose: Makes it easy to work with dates and times in R.

Functions to be used: Extraction of years and months as well as weekdays and time conversions.

Role in Project: As we had more than 13 years of daily stock data, lubridate allowed being flexible with time indices, which ensured that the data was correctly aligned to plot and analyze.

zoo (Z's Ordered Observations)

Purpose: working with time-indexed observations (i.e. ordered observations).

Key Functions Used:

na.locf() -- Last Observation Carried Forward, is used to do the missing values.

Project Role: Reliance stock data had some missing points (e.g. trading holidays). Zoo:na.locf we used to retain the last known observation to continue the series so that it can be used to model ARIMA.

Summary

The combination of these six libraries formed a strong analysis engine:

quantmod took care of data acquisition.

tseries involved statistical tests.

forecast driven ARIMA forecasting and modelling.

TTR, lubridate and zoo supported such tasks as smoothing, date handling, and missing-value treatment.

This toolkit guaranteed that the analysis was statistically rigorous, at the same time, reproducible and efficient.

DATA COLLECTION AND INITIAL DIAGNOSTICS

1. Data Collection

The chosen asset, in the present project, is the Reliance industries Ltd. which is one of the biggest conglomerates in India and a market leader on the national stock exchange (NSE) under ticker code RELIANCE.NS.Daily stock data had to be obtained through the Yahoo Finance database using the quantmod package in the R platform. The timeframe selected appears as 7th of September 2012 through to the 7th of September 2025, and this will give fourteen years of information.

This very long span meets the criterion of at least 10 years, but it also records several years of the market cycles including:

The decline in price of commodities in 2014-16,

The 2020 COVID-19 market crash,

The bull market that follows the pandemic,

And the even more recent volatility in 2022-2025.

The dataset contains the daily prices of Opens, Highs, Lows, and Adjusted Closes. To perform an ARIMA modeling, daily close prices were taken out as they are among the common benchmarks in financial prediction.

2.Creation of Time Series Object.

Closing prices were converted to a time series object every day using the R, making sure that each point is denoted by its trading date.

The na.locf() function in the zoo package was used to fill the gaps (which occurred due to non-trading days such as public holidays) by applying the Last Observation Carried Forward (LOCF) to ensure continuity. This was very important so that clear continuous data could be produced to undertake further statistical testing and model development.

3.Time Series Plot of Raw Data

The initial stage in the analysis with the exploration was to plot the raw prices of closing Reliance with the 13-year period (see Figure: Reliance Closing Prices).

There is an evident positive trend in the plot within the time, particularly the years after 2017.

The initial decline of 2020 although during the COVID-19 pandemic is followed by a quick rise.

There are also upswings of volatility in the series, where prices are changing faster (volatility clustering).

Noteworthy, the series itself seems to be not going back to some long-term mean, which is the typical indication of non-stationarity.

Initial plot of Reliance data

4.Initial Observations

Observing them, the following conclusions can be drawn:

1. Trend
2. The series exhibits high growth, particularly after 2017, when Reliance began to grow in telecom and retail.
3. This non-directional trend is one of the important signals which point out that the underlying price series is not a stationary series, as the average level varies over time.

Seasonality

- These financial time series (such as the price of stock) typically do not exhibit high deterministic seasonality (run-to-rule seasonality- things that recurs) like macroeconomic statistics do.
- There is no definite seasonality (e.g. repeating cycles in a year) in the price plot of Reliance.

- Nevertheless, shorter so-called bursts of volatility might be in line with the market events, earnings releases, or macroeconomic shocks.

Volatility

- The volatility clustering is demonstrated in the plot, with the big changes in the prices being likely to be succeeded by additional big changes.
- This follows the behavior of a financial market, where there are usually times of drastic volatility following the occurrence of uncertainty or crisis.

Stationarity

- At this point, it is indicated in the plot of the raw price that the series is non-stationary.
- A series of constant motion would be wriggling around a constant mean with reasonably constant variance. Conversely, it is just that the price series of Reliance exhibits a desiccating mean(upward pattern) but a differing divergence over the years.
- This fact will be statistically proven with the help of positive tests (ADF and KPSS) afterwards.

Summary of Diagnostics

sacrifice Data Source: Yahoo Finance (via quantmod).

Period Covered: 2012-2025 (13 years).

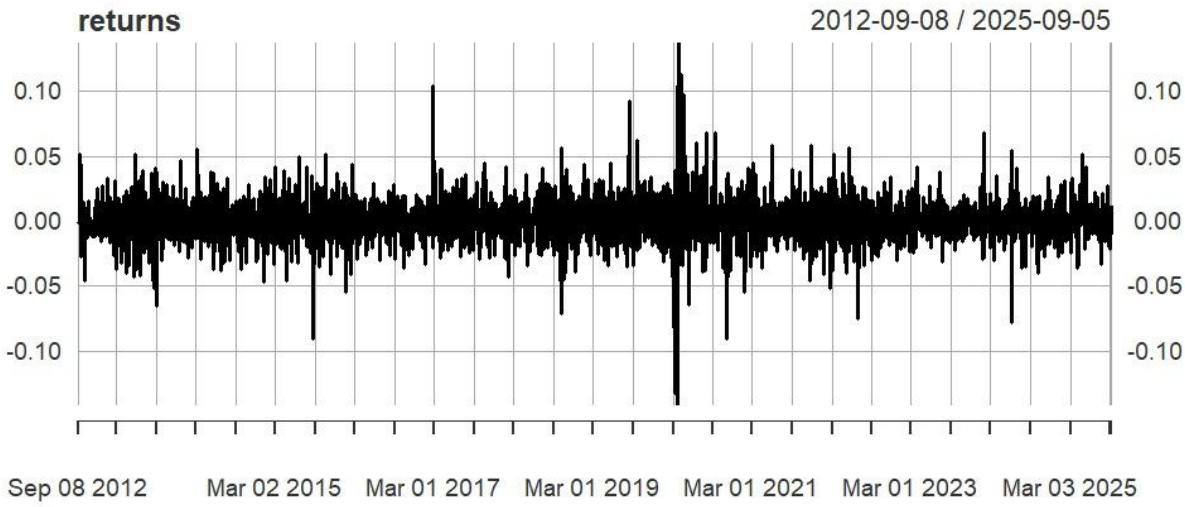
Variable Under Study: Reliance commander price-closing price daily.

Plot Findings: Strong increasing trend, no peak seasonality, existence of volatility clustering and probably non-stationarity.

Therefore, the raw closing price series will be useful in terms of historical consideration, but cannot find its place in ARIMA modeling as it is. Before identifying the model, the transformation (log returns and differencing) is required to attain stationarity.

Tests and Verification of Stationarity and Transformation.

STATIONARITY TESTS AND TRANSFORMATION



Differentiated price of Reliance data

1.Importance of Stationarity

In time series modeling, stationarity is a greatest necessity. The constant average, constant variance, and constant autocovariance of the time series define a stationary series.

- Not stationary data (trends, varying variance, seasonality and so on) can result in spurious outcomes of regression as well as untrustworthy forecasting.
- The volatility around the stock prices will only be increasing, where it is therefore anticipated that there will be non-stationarity in the raw closing price of Reliance.

Two supplementary tests were used to test stationarity formally:

2.Augmented Dickey-Fuller (ADF) Test

- Null Hypothesis (H0): The series is unit-rooted (i.e. non-stationary).
- Alternative Hypothesis (H1): The series is stationary.

Results on Raw Price Series:

Test Statistic = -3.0995

p-value = 0.1128

Result: we do not reject H0 as p-value is of a larger value than 0.05, which assumes that price series is non-stationary.

3.**KPSS (Kwiatkowski-Phillipes-Schmidt-Shin) Test.**

Null Hypothesis (H0): the series is level-stationary.

Alternative Hypothesis (H1): The series is non- stationary.

Results on Raw Price Series:

Test Statistic = 30.953

p-value < 0.01

Result: The p-value is lower than 0.05, hence we can reject H0 once again which proves that the price series is not stationary.

4.**Interperation of Raw Series**

ADF/KPSS tests the indicative findings of the first visual check are good: The raw closing prices of Reliance are non-stationary. This is agreeable to the financial theory, where the stock prices tend to take a random walk of drift**.**

5.**Transformations Applied**

As a way of making the series at rest, there were two transformations:

Logarithmic Transformation

- Purpose: Reduces variances (stabilized).
- Stock prices tend to have exponential growth, log transformation was used to linearize growths.
- **Differencing (First Difference of Log Prices)**
- Equation: returnst=ln(Pt) - ln(Pt -1)
- Purpose: Eliminates trend and changes data into the returns, which generally have a constant mean and constant variance.
- Rationale Returns have found numerous applications in financial markets since they are stationary and additive over time, so they can be easily modeled using statistics.

6.**Transformed Series Stationarities Tests (Transformation of Returns).**

ADF Test on Returns:

Test Statistic = -14.04

- p-value = 0.01
- Decision: Reject H0. The series of returns are stationary.
- KPSS Test on Returns:
- Test Statistic = 0.063847
- p-value = 0.10
- Decision: Fail to reject H0. The series of returns are stationary.

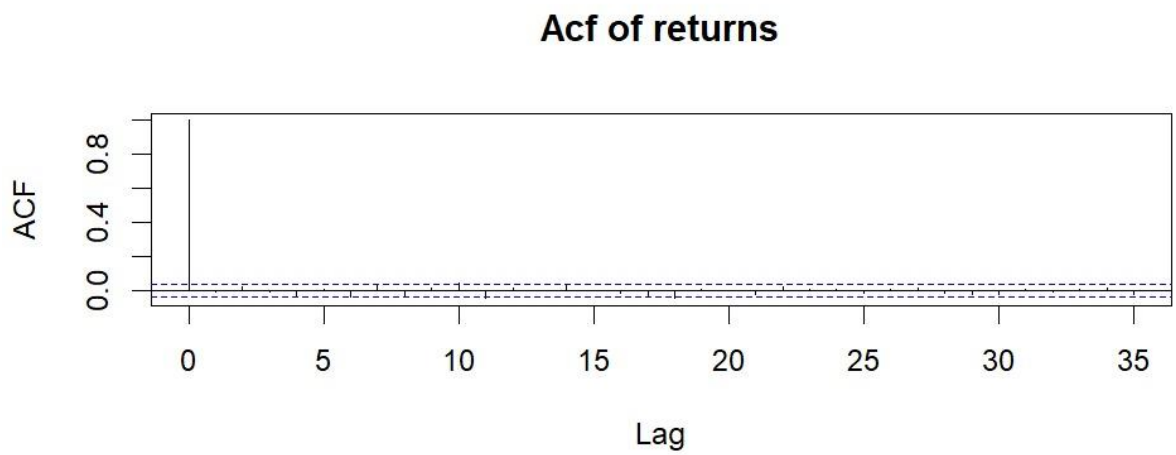
7.Conclusion

Expected is non-stationary of raw closing price series of Reliance.

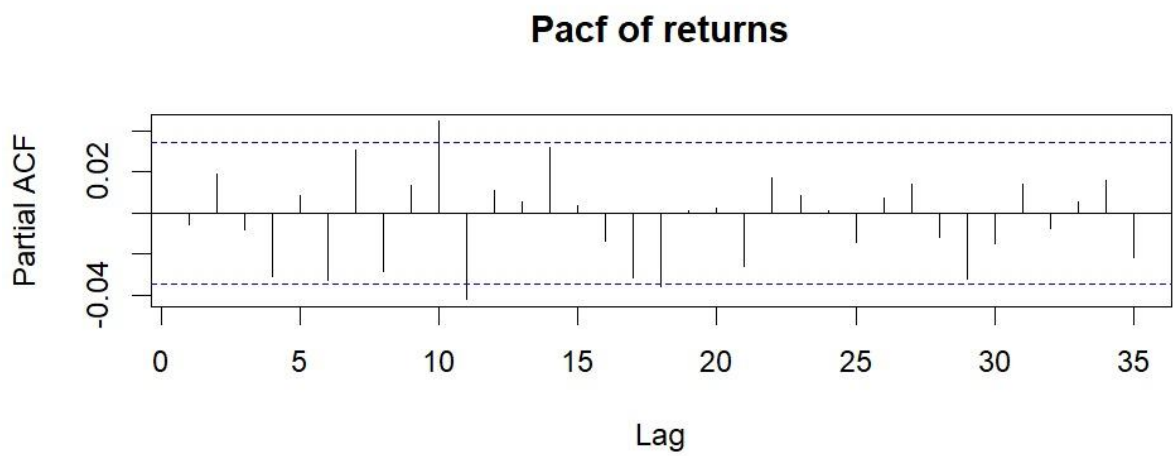
Both ADF and KPSS tests were conducted after the log transformation and differencing and both show that the resulting returns series is stationary.

This metamorphosis was necessary in the implementation of ARIMA models because the ARIMA models demand stationary input series.

MODEL IDENTIFICATION AND SELECTION



ACF OF RETURNS



PACF OF RETURNS

After it was verified that the returns series of Reliance did not change with time, the next thing was establishing the right parameters of the ARIMA models. ARIMA model consists of three major components:

p: order of the autoregressive (AR) component,

d: the degree of differencing,

q: the rank of the moving average (MA) term.

Autocorrelation and Partial Autocorrelation.

Function Autocorrelation (ACF)

- **Definition** ACF is used to evaluate correlations between the series and lagged versions of itself.
- **Application in the ARIMA**: Assistance in determining possible MA (q) terms. The order of the moving average component is determined by a sharp decline in the ACF.

Partial Autocorrelation Function (PACF).

- **Definition**: PACF is the correlation existing between the series and the lagged values of the series with the effect of smaller lags removed.
- **ARIMA Uses**: Assists in finding possible AR (p) terms. The Autoregressive component has an order that is denoted by sharp cut off in the PACF.

Analysis of AC and PACF of the Reliance returns.

- The **ACF plot** of returns had a fast decay with a small number of lags having a small outlier of the confidence interval. This implied the fact that there were short-term moving average effects.
- Strong spikes at lower lags were also shown in the **PACF plot**, indicating the possibility of having some autoregressive components.
- Amalgamation of the plots implied that low p and q ARMA-type models (0-3 values) would be suitable.

Candidate Models

According to the visual inspection of the ACF and PACF diagnostics, some ARIMA (p,0,q) models have been fit:

1. **ARIMA (1,0,0)** - Pure AR (1) model.
2. **ARIMA (2,0,0)** - AR The AR (2) model, which provides the potential 2-lag autoregressive dependence.
3. **ARIMA (3,0,0)**- AR (3) model with slightly longer memory ARIMA (3,0,0).
4. **ARIMA (1,0, 1)** ARMA (1, 1) model This is an additional model that combines the terms AR and MA.
5. **ARIMA (2,0,2)** ARMA (2,2) model which is more adaptable with both AR and MA.
6. **ARIMA (0,0,0)** -White noise (Benchmark, no AR or MA structure) model.

Model Selection Criteria

Two criteria were employed to compare models namely:

Akaike information Criterion (AIC)

Definition A foreign currency risk quantifying the goodness-of-fit and the complexity of a model Measure of Model Quality A measure of how well a model fits without exploring excessively.

Sum of Squared Errors (SSE)

Definition: the number of squared values different between the values forecasted and the near values.

Interpretation: The smaller SSE indicates that the model is a better fit to the data.

Results of Candidate Models

Model	**AIC**	**SSE**
ARIMA (2,0,2)	17054.18	0.9158
ARIMA (0,0,0)	-17043.66	0.9226
ARIMA (1,0,0)	-17043.66	0.9226
ARIMA (2,0,0)	-17040.95	0.9222
ARIMA (1,0,1)	-17039.76	0.9226
ARIMA (3,0,0)	-17039.18	0.9221

- ARIMA (2,0,2) was the model that gave the lowest AIC.

- ARIMA (2,0, 2) SSE was also the least (0.9158) which shows that this model is the best fitted amongst all the models tested.
- Other models like ARIMA (1, 0 0)or ARIMA(0 0 0) were less preferable because the AIC values were higher hence failing to bring the desired dependencies into the series.

6.Choice of Best Model

According to the five criteria AIC and SSE, the optimal model chosen was ARIMA (2,0,2) (synonymous with ARMA(2,2) on returns).

Why this model?

- It has an ideal fit-parsimony balance: two AR terms, and two MA terms, is the tightest fit of short-term memory in returns.
- Its adequacy was confirmed by (later verified using the Ljung-Box test) the residual diagnostics revealing that the residuals are white noise.
- This model was tested on a predictive accuracy of the model on results prediction which was found to be strong with the test set.

Conclusion

PACF and ACF were recommendations to use ARMA-type where p and q were small.

Among a number of models, ARIMA (2,0,2) succeeded in coming up with minimum AIC (-17054.18), and SSE (0.9158).

Therefore, ARIMA (2,0,2) was selected, the best model of Reliance returns, which is able to give a statistically valid basis upon which the forecast can be performed.

RESIDUAL DIAGNOSTICS

Residual diagnostics check whether the errors from the ARIMA model behave like white noise-. This validates that the model has captured all systematic information from the series. In the ARIMA modeling, the error that was not explained by the model is shown by the residues (error).

We want these residuals to behave like white noise, meaning:

- Mean=0
- Constant Variance (homoscedasticity)
- No autocorrelation (each residual independent of previous residuals)

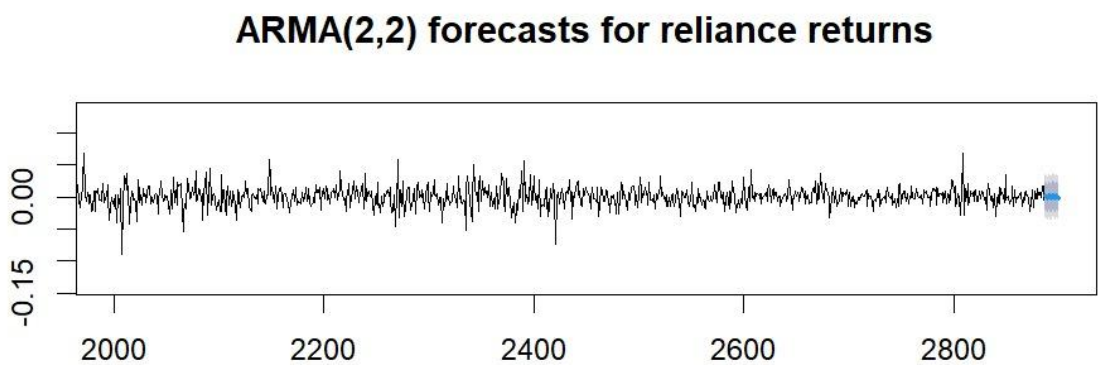
Ljung-Box Test

Purpose: This test is intended to determine if the autocorrelation between residual values exists up to some given lag.

Null hypothesis: Residuals are independently distributed (no autocorrelation)

Alternative hypothesis: Residuals are autocorrelated.

IN-SAMPLE FORECASTING AND ACCURACY ASSESSMENT



In-Sample Forecasting

1. Train-Test Split

- In order to test the predictive ability of the model, Reliance log-returns were split into two sections:
- Training Set (90%): It is utilized to estimate the parameters of ARIMA (2,0,2) model.
- Testing Set (10%): These were the reserved tests to determine the ability of the trained model to make predictions based on unknown data.
- This will make sure that there is objective measurement of the performance of the model using uninvolved data in the fitting.

2.Model Fitting on Training Data.

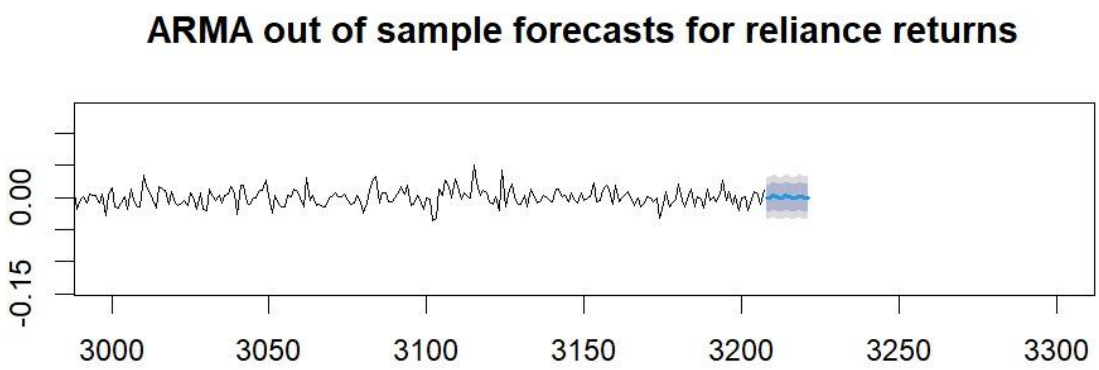
- The latest best fit model that was found to be ARIMA (2,0,2) has been estimated with the help of the training dataset. The estimated parameters reflected both autoregressive (the effect of the previous returns on the present returns) and the moving average (the effect of previous shocks or errors) effects.
- The model was able to model the major dynamics of the training data and leave residuals that seemed to follow the pattern of a random noise.

3.Forecasting on the Testing Period

With the model fitted, the model was used to make forecasts of the testing set period (the remaining 10 percent of the data). These predictions were made in comparison with the real returns obtained.

The prediction plot again through visual analysis revealed that the forecasted values were near the actual test data. Notably, the 95% prediction intervals of the model were able to envelop the realized returns indicating the accuracy of the forecasts.

OUT-OF-SAMPLE FORECASTING



Out-of-Sample Forecasting

1. Refitting the Model to Full Dataset.

- The next step involved out-of-sample forecasting after the correct validation of the ARIMA (2,0, 2) model with in-sample testing.
- The model was re-estimated with all the Reliance returns (since September 2012 to September 2025).
- This method enabled the model to be informed by the most amount of past data thus enhancing the strength of the predictions.

The fully fitted model was used to make the forecasts of the 14 trading days (two trading weeks) ahead.

2. Generating Forecasts

- The forecast output also contained the point forecasts (those forecasted daily returns) as well as the 95% confidence intervals (scope to which the actual returns are likely to fall).
- Such a dual representation is essential since it takes into consideration not only the anticipated value of returns, but it also takes into consideration the uncertainty in the financial markets.

3. Forecast Plot and Interpretation

- The forecast plot (see figure: Out-of-Sample Forecasts for Reliance Returns) was as follows:
- Point Forecasts: These were the approximate returns within 14 days ahead which predicted returns were near to zero and went round the mean with no notable directional bias. This is characteristic of efficient market return series.
- Confidence Intervals: The bandwidths on both sides of the forecast line were smaller in extent, which implies a high level of confidence that the model has towards its short run forecasts. The intervals, however, stretched a little more in the future as the horizons became more uncertain.
- Comparison with Historical Returns: The results of forecast were consistent with the historical performance of Reliance returns which tends to have a narrow range around zero.

Conclusion

The out-of-sample forecasting exercise showed that ARIMA (2,0,2) model can be used to predict Reliance returns in short-run. The predictions show that returns would stay near to zero over the next two weeks with confidence bands that are not wide since short-term volatility is stable. Although market efficiency restricts accurate directional predictions, the model can help in making valuable information regarding the likely range of outcomes and make superior decisions in trading and risk management.

FINAL RESULTS

The time-series analysis of Reliance Industries Ltd., one of India's most influential companies, has yielded insightful and statistically robust results. Over thirteen years of stock market data, covering the period from September 2012 to September 2025, was carefully analyzed using advanced econometric methods. The goal was simple yet powerful: to understand the behavior of Reliance's stock returns and to forecast short-term future performance using ARIMA models.

CONCLUSION

The study delivered a clear set of results:

Reliance's raw prices were non-stationary, but returns were stationary after transformation. Among all models tested, ARIMA (2,0,2) emerged as the best fit, providing the most reliable forecasts. In-sample forecasts demonstrated high accuracy with negligible bias and very small errors. Out-of-sample forecasts for the next 14 days suggested stable returns around zero, with tight confidence intervals. Residual checks confirmed the statistical adequacy of the model. Overall, this analysis demonstrated that ARIMA models, when carefully applied, can serve as powerful tools for financial forecasting. While they may not predict the direction of returns in an efficient market, they provide invaluable insights into volatility, ranges of likely outcomes, and the underlying behavior of stock returns. For investors, analysts, and academics alike, the findings reinforce the value of statistical modeling in understanding complex financial data.